Manual Calculation

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1. Program A : 8-bit Addition & Subtraction

given A = 5AH , B = 27 H

Addition :

5AH + 27H = 81H

Sum = 81H, Cy = 0

AC = 1, P = 1, Z = 0, S = 1

Flag byte = 94H

Signed Overflow = Yes

Subtraction :

5AH − 27H = 33H

Difference = 33H, Borrow = 0

AC = 0, P = 1, Z = 0, S = 0

Flag byte = 06 H

Signed Overflow = No

2. Program B : 16-bit Addition and Subtraction

given X = 1234H , Y = 4321 H

Addition :

1234 H + 4321 H = 5555 H

carry = 0

Subtraction :

1234H − 4321 H = CF13H

Borrow = 1

3. Program C : 8-bit Multiplication and division

given Multiplication :

14H × 06 H = 0078 H

Product = 0078H

Division

F2H ÷ 0A H

Quotient = 18H

Remainder = 02H

verification :

F2H = (18H × 0AH) + 02 H

4. Program D : 16-bit Multiplication and division

given,

Multiplication :

1357H × 0025 H = 0002 CA13H

upper word = 0002 H

lower word = CA13H

Division:

9A23H ÷ 0011H

Quotient = 0911H

Remainder = 0002H

verification:

9A23H = (0911H X 0011H) + 0002H